Unit 7, Lesson 3: Adding Integers Using Integer Chips

Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

Learning Targets

- ☐ I can express integer addition using integer chips.
- ☐ I can add integers using integer chips.

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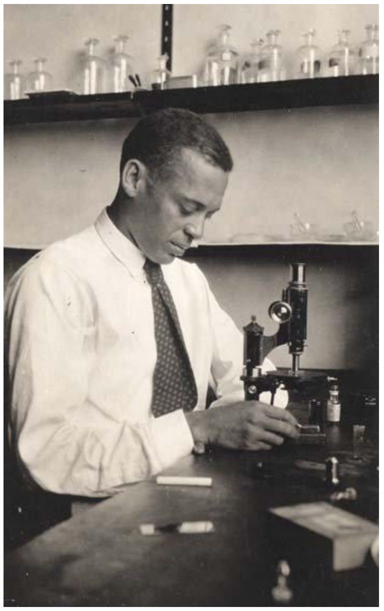
7.3.1 Warm-Up: Below Sea Level



Dr. Ernest Everett Just was a pioneering African American biologist who conducted research at Woods Hole Marine Biological Laboratory in Massachusetts. He published two books and more than 70 scientific papers, most of which focused on the cell. Today, marine biologists use submarines to study marine life.

A submarine explorer begins at sea level and descends at a rate of 150 feet per minute.

1. Sketch a picture that shows how far below sea level the submarine will be in 3 minutes.

2. Add a bird that is flying $1{,}000$ feet above sea level to your picture.

3. What number represents sea level?

4. Complete the statement.

   The location of the submarine is represented by a

   - ○ positive value,
   - ○ negative value,

   while the location of the bird is represented by a

   - ○ positive value.
   - ○ negative value.

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7.3.2 Guided Instruction: Zero Pairs

Positive and negative values are often useful in the contexts of money, banking, and accounting.

- In the context of money, [○ positive / ○ negative] numbers can be used to represent money spent or money owed to someone.
- [○ Positive / ○ Negative] numbers can be used to represent money earned.

Money can be used to learn how to add integers.

- A value of $+1$ means that you ________________ $\$1$.
- A value of (-1) means that you ________________ $\$1$ or __________ $\$1$.

The table below shows a positive integer chip, which represents earning $\$1$, and a negative integer chip, which represents spending $\$1$.

Earn	Spend
	

Integer chips can also be used to represent integers.

Integers are whole numbers and their opposites.

A positive number can be written with a positive sign, $+1$ or $(+1)$, or it can be written without any sign, 1.

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A negative number can be written as -1 , (-1) or $^{-}1$.

1. Complete the table. In the last column, sketch the integer chips that represent the phrase and the integer. Place a + sign in a circle to represent  and place a — sign in a circle to represent .

Phrase	Integer	Integer Chips
Earned $6	$(+6)$	
Spent $3	-3	

2. Complete the table by sketching integer chips for each phrase.

Earned $4	Spent $4	Earned $4, Then Spent $4

3. Joleen earned $4, then spent $4. How much money does she have left?

The last column in the table and problem above both represent a ***zero pair***.

A ***zero pair*** is a pair of numbers whose sum is zero.

4. Complete the table by drawing the negative integer chips that create a zero pair. In the **sum** column, draw all of the chips from both columns.

Positive	Negative	Sum
		

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Each zero pair can be represented by an equation. The box shows two mathematical equations that represent the zero pair of  and .

Zero Pair as a Mathematical Equation

$^{+}1 + (-1) = 0$ or $1 + {}^{-}1 = 0$

5. Write an equation that represents the **sum** column in the previous table.

6. Complete the table by drawing the positive integer chips that create a zero pair. In the **sum** column, draw all of the chips from both columns.

Positive	Negative	Sum
	 	

7. Write an equation that represents the **sum** column in the previous table.

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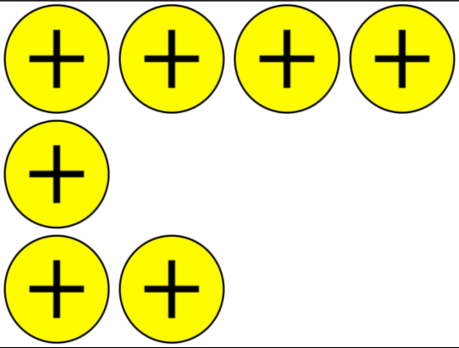
7.3.3 Exploration: Adding Integers with the Same Sign

1. Work with your partner to complete the table.

Mathematical Expression	Money Sentence	Integer Chips	Sum
	I earn $\$5$ babysitting and another $\$2$ for cleaning the dishes.		
$3 + 2$			
$^{-}4 + {}^{-}3$			
$(-2) + (-1)$			

2. With your partner, discuss the patterns you notice about the signs of the sums when adding integers with the same sign.

3. Were any zero pairs created when adding integers with the same sign? Explain.

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7.3.4 Bring It Together: Adding Integers with the Same Sign

1. Complete the visual model of the rule of adding two positive integers by drawing the integer chip that represents the **sign** of the sum.

Visual Model of Addition	Sign of Sum
	

2. Complete the visual model of the rule of adding two negative integers by drawing the integer chip that represent the **sign** of the sum.

Visual Model of Addition	Sign of Sum
	

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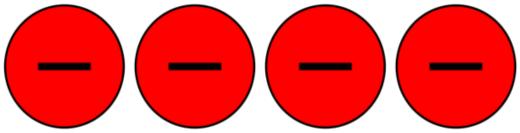
7.3.5 Exploration: Using Zero Pairs to Add Integers with Opposite Signs

It is possible to add a positive number and a negative number. To find the answer using integer chips, combine a  and  to create a zero pair. Make as many zero pairs as possible. Then, determine the number of chips that are left over.

1. Complete the table to show how the sum $5 + ({}^{-}4)$ can be represented using integer chips.

Step	Response
Represent the first addend with integer chips.	
Represent the second addend with integer chips.	
Determine the number of zero pairs created, if any.	
Cross out the integer chips that represent zero pairs to indicate they are eliminated from the model.	
Determine the **sum** using the number of integer chips remaining.	$5 + ({}^{-}4) =$ _____

2. What do you notice about the remaining integer chips and the sign of the sum?

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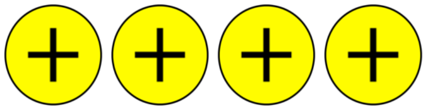
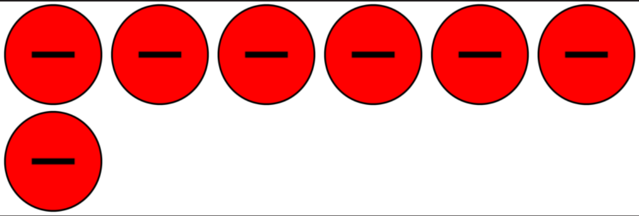
3. Complete the table independently to show how the sum $4+(-7)$ can be represented using integer chips. Check your work with your partner.

Step	Response
Represent the first addend with integer chips.	
Represent the second addend with integer chips.	
Determine the number of zero pairs created, if any.	
Cross out the integer chips that represent zero pairs to indicate they are eliminated from the model.	
Determine the **sum** using the number of integer chips remaining.	$4+(-7)=$ _____

4. What do you notice about the remaining integer chips and the sign of the sum?

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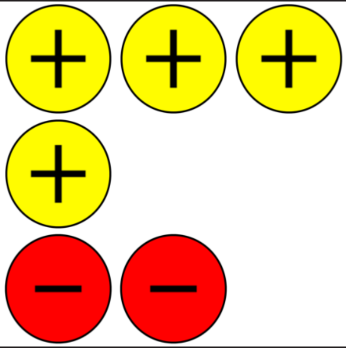
5. Work with your partner to complete the table. The first row is completed.

Mathematical Expression	Money Sentence	Integer Chips	Sign of Sum	Sum
$4 + (-2)$	I earn $\$4$ and spend $\$2$.		$+$	$+2$
$3 + (-6)$				
$^{-}7 + 5$				
$(-1) + 8$				

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6. Complete the table below independently. Check your work with your partner.

Mathematical Expression	Number of Zero Pairs	Sign of Sum	Sum
$(-10) + 12$	10	$+$	2
$12 + -7$			
$(-11) + 4$			
$8 + {}^{-}8$			
${}^{-}10 + 3$			
$9 + (-2)$			

7. What do you notice about expressions where the sum is a positive number?

8. What do you notice about expressions where the sum is a negative number?

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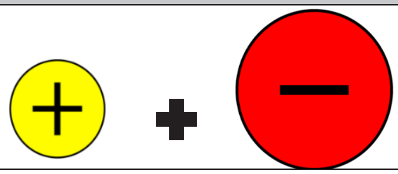
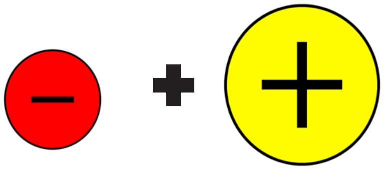
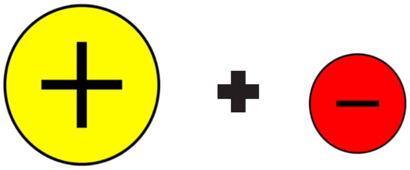
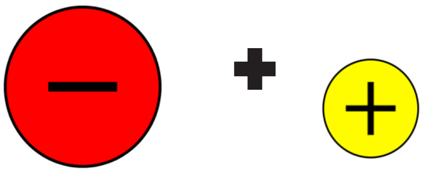
7.3.6 Bring It Together: Adding Integers with Opposite Signs

The table below shows visual models of rules for adding integers with opposite signs. In each visual model, the larger circle symbolizes the integer represented by the most integer chips.

1. Write an example of the rule. The first row has been completed.

Visual Model of Addition	Sign of Sum	Example
		$2 + (-5) = -3$
		
		
		

2. Give two observations about the rules for adding integers with opposite signs.

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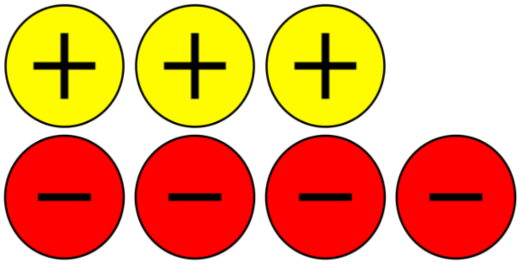
7.3.7 Wrap-Up: Ticket Out the Door

1. Simone created the sketch below to represent the expression $4+(-3)$.



Her sketch is incorrect. Explain why and create a new sketch to find the sum.

2. Complete the table for the expression below.

Expression	$3+{}^{-}5$
Integer Chips	
Money Sentence	
Sign of the Sum	
Sum	

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